

# Closed, Open Source, and Open Weight AI Models

Understanding the three levels of model access — and why the distinction matters

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AI model releases fall on a spectrum from fully closed to fully open. The three terms used most often — **closed (proprietary)**, **open weight**, and **open source** — describe how much of a model's underlying components a developer makes available: just the ability to use it, the trained parameters themselves, or everything needed to rebuild it from scratch.

## The Three Categories

### 1. Closed / Proprietary Models

The model's weights, training code, and training data all stay private. Access is provided only through a paid API or hosted product, under license terms set by the company. Users cannot download, inspect, or run the model on their own infrastructure.

- **Examples:** OpenAI's GPT models, Google's Gemini, Anthropic's Claude.
- **Pros:** Usually the most capable/frontier performance; provider handles hosting, safety tuning, updates, and support.
- **Cons:** No transparency into training data or weights; usage depends on the provider's pricing, availability, and policies; data typically leaves your infrastructure.

### 2. Open Weight Models

The trained model parameters ("weights") are published for anyone to download, run, and fine-tune — but the training code, full training data, and complete methodology are not released. You get the finished model, not the recipe used to build it.

- **Examples:** Meta's Llama models, Alibaba's Qwen, DeepSeek's V/R series, Google's Gemma, Moonshot's Kimi, Zhipu's GLM, OpenAI's gpt-oss.
- **Pros:** Can be self-hosted, fine-tuned, and run offline or air-gapped; often free or low-cost; increasingly close to closed-model performance on many tasks.
- **Cons:** Cannot fully audit how the model was trained or what data it saw; licenses vary widely — some (e.g., Meta's Community License) impose usage restrictions above certain scale, which is why organizations such as the Open Source Initiative and Free Software Foundation do not consider them truly open source.
- This is by far the most common category for what the industry casually calls "open source" models today.

### 3. Open Source Models

Everything needed to reproduce the model from scratch is released: weights, training code, and enough detail about the training data and methodology for others to audit or rebuild it, under a permissive license with few or no usage restrictions. The Open Source Initiative formalized this definition for AI in 2024.

- **Examples:** Research-driven projects such as AI2's OLMo and BLOOM. Very few commercially backed "open" models fully qualify.

- **Pros:** Maximum transparency and reproducibility; supports independent safety research and auditing; no permission needed for any use.
- **Cons:** Rare in practice — releasing full training data at scale is costly and sometimes legally complicated, so most companies stop at open-weight.

## Side-by-Side Comparison

| Aspect                   | Closed / Proprietary        | Open Weight                                                               | Open Source                                      |
|--------------------------|-----------------------------|---------------------------------------------------------------------------|--------------------------------------------------|
| Weights available?       | No — API access only        | Yes — downloadable                                                        | Yes — downloadable                               |
| Training code available? | No                          | Usually no                                                                | Yes                                              |
| Training data disclosed? | No                          | Rarely, partially at most                                                 | Yes (or close to it)                             |
| Can self-host?           | No                          | Yes                                                                       | Yes                                              |
| Can fully audit/rebuild? | No                          | No                                                                        | Yes                                              |
| Typical license          | Commercial terms of service | Custom or permissive (Apache 2.0, MIT, or restrictive community licenses) | Permissive open-source license (Apache 2.0, MIT) |
| Representative examples  | GPT-5.x, Gemini, Claude     | Llama, Qwen, DeepSeek, Gemma, GLM, gpt-oss                                | OLMo, BLOOM                                      |

## Why the Distinction Matters

- **Vendor lock-in and cost:** closed models require ongoing API spend and dependence on one provider; open-weight models can be self-hosted and customized at fixed infrastructure cost.
- **Data privacy and control:** open-weight models can run fully offline or air-gapped, which matters for regulated or sensitive data; closed models require sending data to the provider (though enterprise tiers often add contractual protections).
- **Transparency and safety research:** only open-source releases let independent researchers fully audit training data and methodology for bias, safety, or copyright concerns; open-weight releases allow behavioral testing but not full provenance auditing.
- **Licensing risk:** "open" branding does not guarantee unrestricted use — always check the specific license (e.g., Meta's Llama Community License includes a usage cap for very large companies) before deploying commercially.

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Sources: Open Source Initiative's 2024 definition of Open Source AI; PBS News / The Conversation explainer on closed, open-source, and open-weight AI; Microsoft's "Open Weights and American AI Leadership" statement; industry model-license trackers (Telnyx, Faros, GEO Toolbox, TechRound) reviewed for current model classifications as of mid-2026.

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